

Assignment : 1

Date : 16/07/2026

Session : 1

❖ **Name 5 IT job roles and write one line about what each role does.**

- 1. Software Developer** – Writes and maintains software applications.
- 2. Software Tester (QA)** – Tests software to find bugs and ensure quality.
- 3. UI/UX Designer** – Designs user-friendly interfaces and user experience.
- 4. Data Analyst** – Analyzes data to help businesses make decisions.
- 5. AI Engineer** – Builds AI and Machine Learning models.

❖ **List 3 product companies and 3 service companies with one example of work each does**

➤ **Product Companies :-**

- 1. Google** – Builds products like Gmail, Google Search, Android.
- 2. Microsoft** – Develops Windows, Office, Azure.
- 3. Zoho** – Creates CRM, Mail, Books, and business software.

➤ **Service Companies :-**

- 1. TCS** – Provides software development and IT services.
- 2. Infosys** – Offers IT consulting and software solutions.
- 3. Wipro** – Delivers IT services, cloud, and business solutions.

❖ **Write the full forms and one-line use of: GitHub, Slack, Jira, IDE, API.**

1. GitHub –

Full Form: Git + Hub (No official expansion)

Use: GitHub is a cloud-based platform used for storing, managing, collaborating, and tracking source code using Git version control.

2. Slack –

Full Form: Searchable Log of All Communication and Knowledge

Use: Slack is a team communication platform used for messaging, file sharing, voice/video meetings, and project collaboration.

3. Jira –

Full Form: Jira does not have an official full form (named after the Japanese word "Gojira").

Use: Jira is a project management and issue-tracking tool used for planning, assigning, and tracking software development tasks.

4. IDE –

Full Form: Integrated Development Environment

Use: An IDE is software that provides a code editor, compiler/interpreter, debugger, and other tools to help developers write, test, and debug programs.

5. API –

Full Form: Application Programming Interface

Use: An API allows different software applications to communicate and exchange data with each other securely and efficiently.

❖ **Name any 5 programming languages and mention one popular use case for each**

1. **Java** – Java is commonly used for Enterprise Applications, Android App Development, and Banking Software..
2. **Python** – Python is widely used for Artificial Intelligence, Machine Learning, Data Science, Automation, and Web Development..
3. **C++** – C++ is used for Game Development, Operating Systems, Embedded Systems, and High-Performance Applications.
4. **JavaScript** – JavaScript is mainly used to build interactive websites and modern web applications.
5. **PHP** – PHP is used for developing dynamic websites, content management systems like WordPress, and web applications.

❖ **What is the difference between a developer and a tester?**

Answer in 2 lines

- **Developer:** A Developer writes, designs, and maintains software by implementing new features, fixing bugs, and improving application performance.
- **Tester:** A Tester verifies the software by finding bugs, checking functionality, validating requirements, and ensuring the application meets quality standards before release.

Session : 2

❖ What are the three main layers of software architecture?

Write one line about each.

1. **Presentation Layer (UI)** – The Presentation Layer, also called the User Interface (UI), is the part of the application that users interact with; it displays information, accepts user input, and sends requests to the Business Logic Layer
2. **Business Logic Layer** – The Business Logic Layer processes user requests, applies business rules, performs calculations, validates data, and communicates between the Presentation Layer and the Database Layer.
3. **Database Layer** – The Database Layer stores, manages, retrieves, updates, and deletes application data using database management systems such as MySQL, PostgreSQL, or MongoDB.

❖ Define client and server in your own words with one real-life example.

- **Client** – A Client is a device or software application, such as a web browser or mobile app, that sends requests to a server to access information or services over a network.
- **Server** – A Server is a powerful computer or software system that receives requests from clients, processes those requests, accesses the required data, and sends the response back to the client.
- **Real-Life Example** – When you open Instagram, the Instagram mobile app acts as the client and sends a request to Instagram's server for your latest feed. The server retrieves the required posts from its database and sends them back to the app, where they are displayed on your screen.

❖ **Write 2 differences each between desktop apps, web apps, and mobile apps.**

1. Desktop Application

- A desktop application is installed directly on a computer or laptop and usually runs using the computer's operating system.
- Most web applications require an internet connection because they communicate with a remote server.

2. Mobile Application

- A mobile application is installed on smartphones or tablets through app stores such as Google Play Store or Apple App Store.
- Mobile applications can access device features such as the camera, GPS, fingerprint sensor, notifications, microphone, and contacts.

3. Web Application

- A web application runs inside a web browser such as Chrome, Edge, or Firefox and does not require installation.
- Most web applications require an internet connection because they communicate with a remote server.

❖ **Name any 3 databases and mention whether each is SQL or NoSQL.**

Database	Type
MySQL	SQL
PostgreSQL	SQL
MongoDB	NoSQL

Session : 3

❖ Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP

1. HTTP – HyperText Transfer Protocol

- HTTP is a communication protocol used to transfer web pages and data between a web browser (client) and a web server over the Internet.

2. HTTPS – HyperText Transfer Protocol Secure

- HTTPS is the secure version of HTTP that encrypts data using SSL/TLS, protecting communication between users and websites.

3. FTP – File Transfer Protocol

- FTP is used to upload, download, and manage files between a local computer and a remote server over a network.

4. SMTP – Simple Mail Transfer Protocol

- SMTP is the standard protocol used to send emails from an email client to a mail server and between mail servers.

❖ What is the difference between IPv4 and IPv6? Answer in 2 lines.

1. **IPv4 :-** IPv4 uses a 32-bit address, supports approximately 4.3 billion unique IP addresses, and is written in decimal format (Example: 192.168.1.1).
2. **IPv6 :-** IPv6 uses a **128-bit address**, supports an almost unlimited number of unique IP addresses, and is written in hexadecimal format (Example: 2001:db8::1).

❖ **Run nslookup google.com on your system and write down the IP address you get.**

➤ **IP :- 77.77.77.77**

➤ **Address :- 10.19.3*.27**

❖ **What do these HTTP status codes mean: 200, 301, 404, 500, 403?**

- 1. 200 :- OK** – The request was successful, and the server returned the requested resource.
- 2. 301 :- Moved Permanently** – The requested resource has been permanently moved to a new URL, and browsers should use the new address.
- 3. 404 :- Forbidden** – The server understands the request but refuses to allow access because the user does not have the required permissions.
- 4. 500 :- Not Found** – The requested page or resource could not be found on the server.
- 5. 403 :- Internal Server Error** – The server encountered an unexpected error while processing the request.

❖ **Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example**

Explain DNS hierarchy (root, TLD, domain, subdomain)
with www.google.com as an example

- 1. Root (.) :-** The Root is the highest level of the DNS hierarchy and directs queries to the appropriate Top-Level Domain (TLD) servers.

- 2. Top-Level Domain (TLD) :-** The .com is the Top-Level Domain that identifies the domain category.
- 3. Domain :-** The google part is the registered domain name owned by Google.
- 4. Subdomain :-** The www is a subdomain used to access Google's web service.

Session : 4

❖ List all 6 phases of SDLC in order with one line about each.

- 1. Requirement Analysis :-** In this phase, the development team gathers and analyzes the client's requirements to understand what the software should do and what features are needed.
- 2. System Design :-** The system architecture, database structure, user interface, and overall software design are planned before development begins.
- 3. Development (Implementation) :-** Developers write the source code, create application features, and implement the design using programming languages and frameworks.
- 4. Testing :-** Testers verify the software by finding bugs, checking functionality, security, and performance to ensure the application works correctly.
- 5. Deployment :-** The completed application is released to users by installing or publishing it on servers, cloud platforms, or app stores.
- 6. Maintenance :-** After deployment, developers fix bugs, improve performance, add new features, and keep the software updated according to user feedback.

❖ Write 3 differences between Agile and Waterfall models.

Agile	Waterfall
Development is done in small iterations called sprints.	Development follows a fixed sequential process from start to finish.
Changes in requirements can be easily accommodated during development.	Changes are difficult and expensive after development begins.
Customer feedback is collected frequently throughout the project.	Customer feedback is usually taken after the complete project is finished.

❖ **Name the 3 main Scrum roles and write one line about the responsibility of each.**

- 1. Product Owner (PO) :-** The Product Owner defines product requirements, prioritizes the backlog, and ensures the team builds features that deliver maximum business value.
- 2. Scrum Master :-** The Scrum Master guides the Scrum process, removes obstacles, facilitates meetings, and helps the team follow Agile principles.
- 3. Development Team :-** The Development Team designs, develops, tests, and delivers working software during each sprint.

❖ **What is a Sprint in Scrum? Mention its typical duration.**

- A **Sprint** is a fixed time period in Scrum during which the development team completes a set of planned tasks and delivers a working product increment.
- **Typical Duration: 1 to 4 weeks**, with **2 weeks** being the most commonly used sprint length.

❖ **What do these Jira board columns mean: To Do, In Progress, Done, Backlog?**

- **Backlog :-** The Backlog contains all project requirements, user stories, tasks, and bug reports that are planned but not yet started.
- **To Do :-** The To Do column contains tasks that have been selected for the current sprint but have not yet been started.
- **In Progress :-** The In Progress column contains tasks that developers are currently working on.
- **Done :-** The Done column contains tasks that have been completed, tested, and are ready for delivery or have already been delivered.

Session : 5

❖ Write the Git command for each: initialize a repo, stage all files, commit with a message, push to remote, pull latest changes.

➤ **Initialize a Git Repository**

`</> bash – git init`

➤ **git init**

`</> bash – git add .`

➤ **Commit with a Message**

`</> bash – git commit -m "Initial commit"`

➤ **Push to Remote Repository**

`</> bash – git push origin main`

➤ **Pull the Latest Changes**

`</> bash – git pull origin main`

❖ **What is the difference between git fetch and git pull? Answer in 2 lines.**

➤ **git fetch :-** `git fetch` downloads the latest changes from the remote repository but **does not merge** them into your current branch.

➤ **git pull :-** `git pull` downloads the latest changes **and automatically merges** them into your current branch.

❖ **Create a new GitHub repository, push any small text file from your local system, and share the repo link.**

➤ **Git Commands :-**

- ✓ `git init`
- ✓ `git add .`
- ✓ `git commit -m "Top_Technology"`
- ✓ `git remote add origin https://github.com/JesilSardhara/Top_Technology.git`
- ✓ `git push -u origin main`

❖ **What is the difference between a Fork and a Pull Request? Answer in 2 lines.**

- **Fork :-** A **Fork** creates a personal copy of someone else's GitHub repository under your own GitHub account so you can make changes independently.
- **Pull Request (PR) :-** A **Pull Request** is a request to merge your changes from a branch or fork into the original repository after review.

❖ **Create a new branch named feature-test, make one commit on it, and merge it back to the main branch. Write the commands you used.**

➤ **Step 1: Create and Switch to a New Branch**

- ✓ `</> bash – git checkout -b feature-test`

➤ **Step 2: Make Changes and Stage Them**

- ✓ `</> bash – git add .`

➤ **Step 3: Commit the Changes**

- ✓ `</> bash – git commit -m "first commit"`

➤ **Step 4: Switch Back to Main Branch**

- ✓ `</> bash – git checkout main`

➤ **Step 5: Merge the Branch**

✓ `bash` – `git merge feature-test`

➤ **Step 6: Push the Updated Main Branch**

✓ `bash` – `git remote add origin`

`https://github.com/JesilSardhara/Top_Technology.git`